Smart Control Tower · Case Study

Amber Press Shop · IoT Daily Performance Review & Deterministic AI Intelligence

AI-Powered Plant Intelligence

Axiom: App Calculates, AI Explains

90m → 10s	0% Errors	100% Private	99.9% Uptime
Review Prep Time	Zero Math Hallucinations	In-Browser Client RAM Only	3-Tier LLM Fallback Chain

The Challenge (Before)

Manual Friction

Plant monitors display live DFOS screens, but daily operational reviews required tedious manual synthesis across 3 separate workbooks:

- **Fragmented Files:** Production, Detailed Downtime, and Scrap Rejections lived in disconnected sheets.
- **Arithmetic Glitches:** 22 rows had OEE > 100% (up to 209%), and downtime fields failed mathematical reconciliation.
- **Delayed Root-Cause Analysis:** Supervisors spent 90+ minutes copy-pasting numbers instead of investigating shop-floor fixes.

The Intervention (After)

Smart Automation

A zero-install, decoupled workspace unifying all 3 reports with instant calculations and bounded AI explanation:

- **Unified Ingestion:** In-browser Drag & Drop harmonizes files by Date, Machine Cell, Model, and Shift.
- **Deterministic Paretos:** Code automatically computes Top 10 Downtime & Scrap rankings before AI sees it.
- **Closed-Loop Actions:** Observed losses link directly to RCA cases, assigned owners, and 30-day recurrence audits.

5-STAGE LOGICAL WORKFLOW PIPELINE

End-to-End Architecture

Stage 1
Ingest
Drop-zone reads Production, Downtime & Rejection XLSX files directly in browser RAM.

Stage 2
Compute
JavaScript calculates Attainment %, OEE, and Top 10 Paretos. Zero server math.

Stage 3
Token Bridge
Python bridge (`127.0.0.1` / Vercel API) validates 32-byte secret tokens & Origin.

Stage 4
Multi-LLM
Cascading fallback: **OpenAI** (1st) → **Mistral** (2nd) → **Groq** (3rd). `store:false`.

Stage 5
Deliver
UI renders sanitized Markdown cards alongside verified mathematical evidence tables.

The Core Engineering Axiom

> "App Calculates, AI Explains"

The LLM is strictly prohibited from doing arithmetic, inventing baselines, or hallucinating causes. Code computes verified facts; AI suggests qualitative engineering checks.

Air-Gapped Rows · Math Integrity Shield · Ephemeral Session

AI Assistant & Diagnostics



- **Monthly & Daily Briefings:** 1-click plain-English executive loss summary.
- **Interactive 'Ask AI':** Plain-language chat (*"Why did Cell 1 stop on Dec 1?"*).
- **Grounded Verification:** Every suggestion cites exact machine dates & hours.